

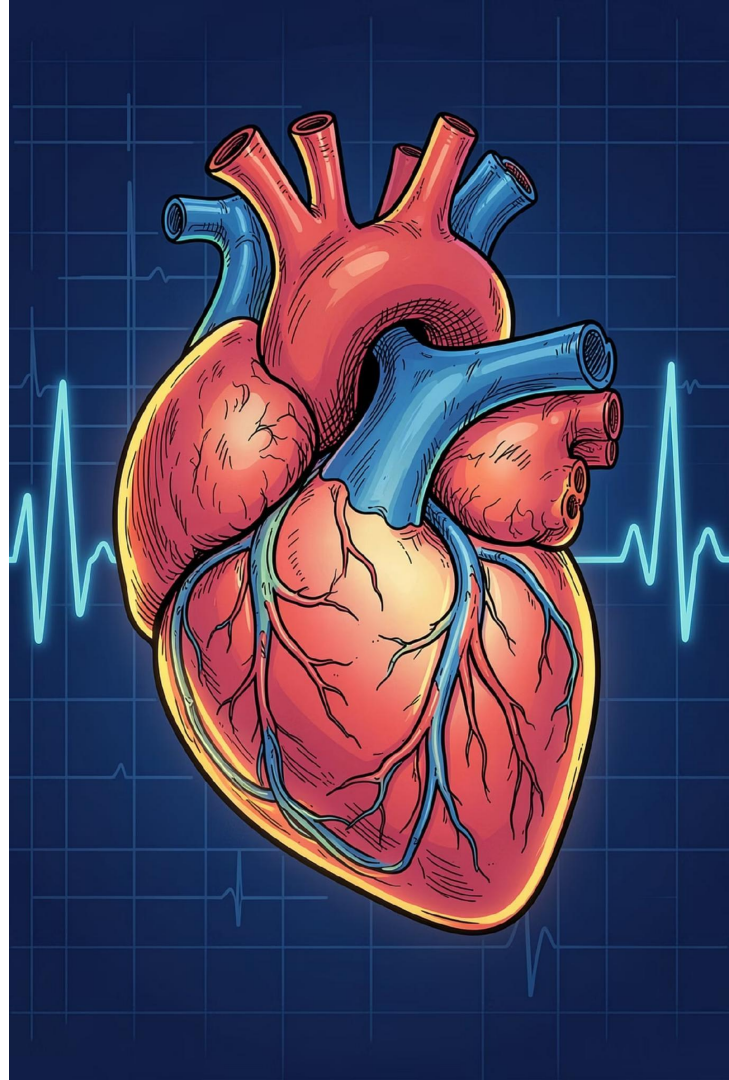
Predicting Heart Disease with Machine Learning

CPE393 — Introduction to Data Science with Python · Summer
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A High-Stakes Classification Problem

Cardiovascular diseases are the leading cause of death worldwide. Early detection is difficult, many clinical measurements are taken during routine checkups, but their combined predictive value is not always clear.

Problem

Can we reliably predict heart disease from standard clinical measurements?

Approach

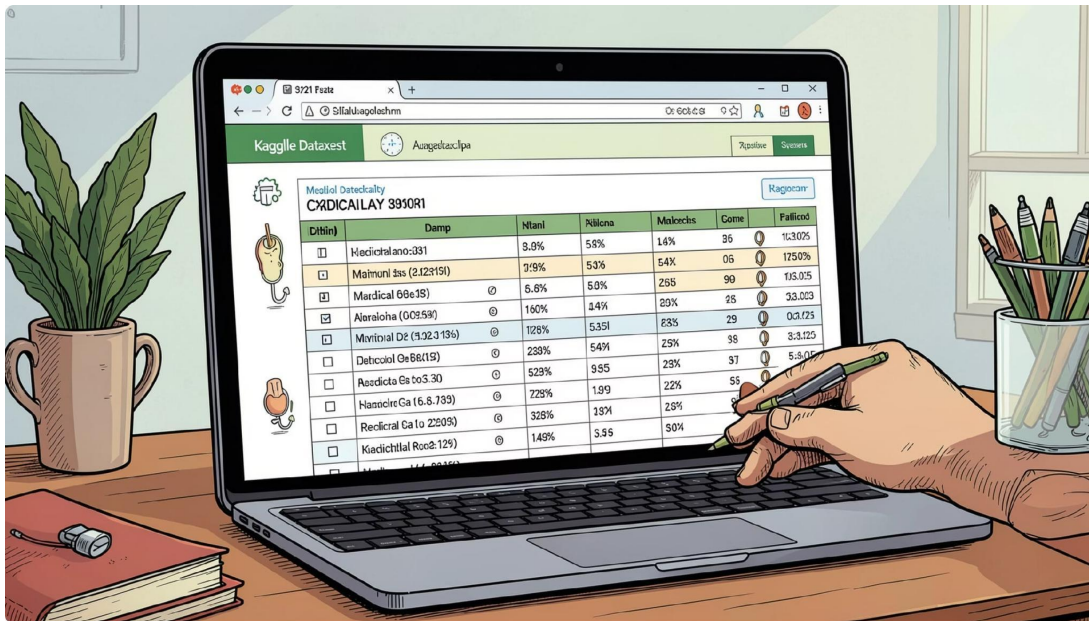
Compare 4 classifiers on a cleaned UCI-derived dataset

Key Insight

Exercise-related features (max heart rate, chest pain) are the strongest predictors



The Heart Disease Dataset

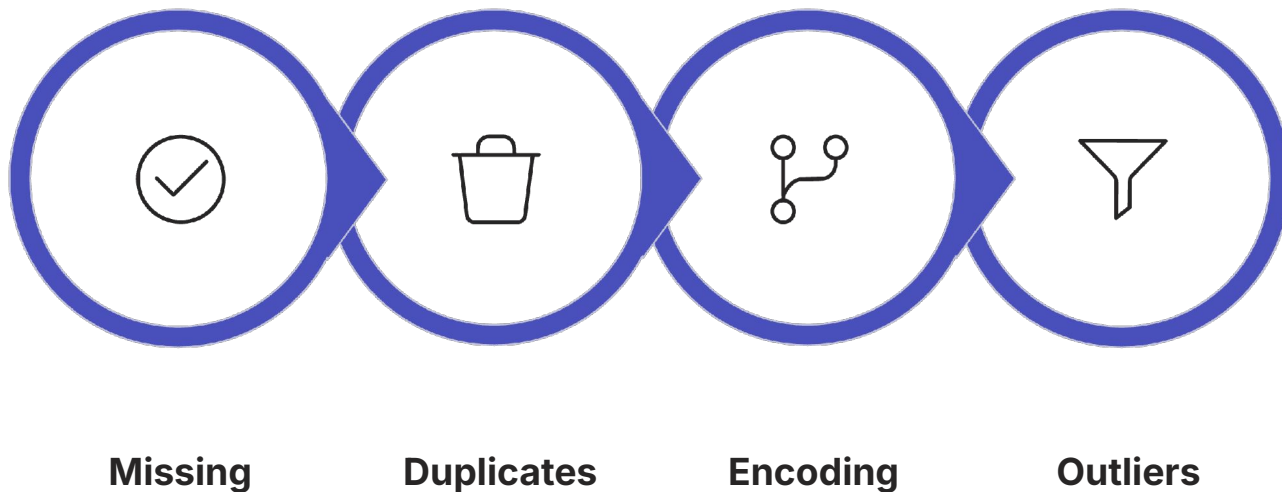


Source & Structure

Derived from the UCI Cleveland database, hosted on Kaggle. Originally 1,025 rows — but **70% were exact duplicates**. After rigorous cleaning: **296 unique patient records**, 14 columns.

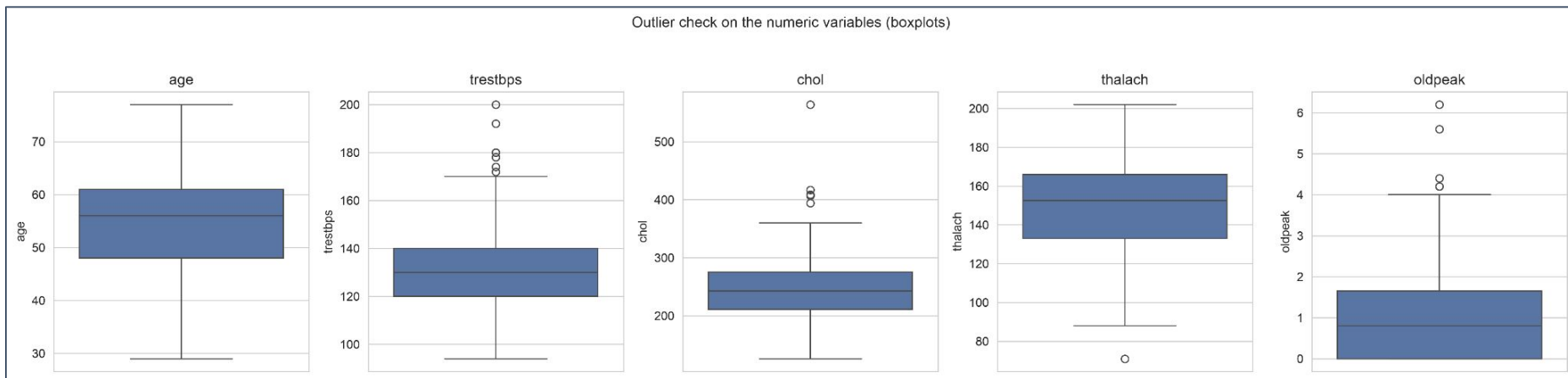
- **Demographics:** age, sex
- **Clinical at rest:** blood pressure, cholesterol, ECG, fasting sugar
- **Exercise-related:** max heart rate, angina, ST depression
- **Target:** 0 = no disease, 1 = disease present

Data Cleaning Pipeline



Cleaning reduced the dataset from 1,025 to **296 unique patients**. Removing duplicates before any train-test split was essential to prevent data leakage and ensure honest model evaluation.

Data Cleaning Pipeline

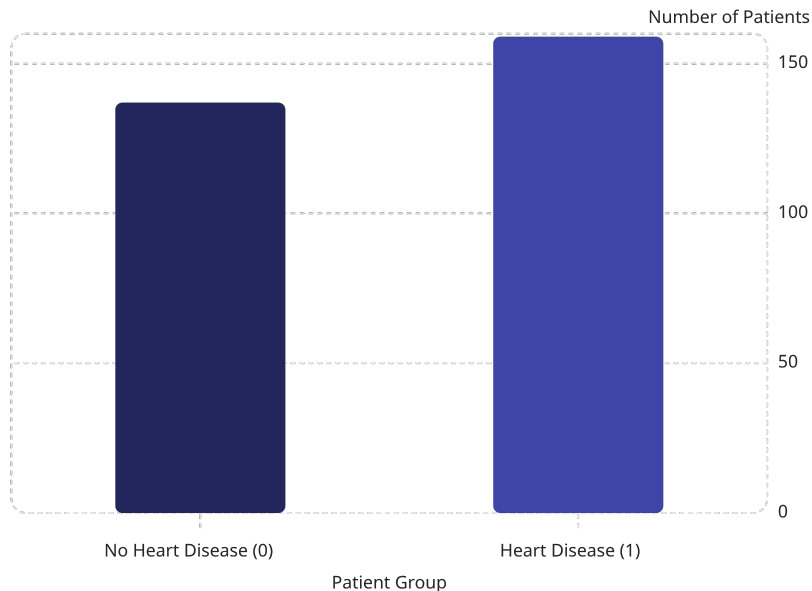


An outlier is a value that sits far outside the typical range of a variable — unusually large or small compared to the rest of the distribution

chol → 5 outliers (IQR method), bounds = [114.6, 371.6]

Exploratory Data Analysis

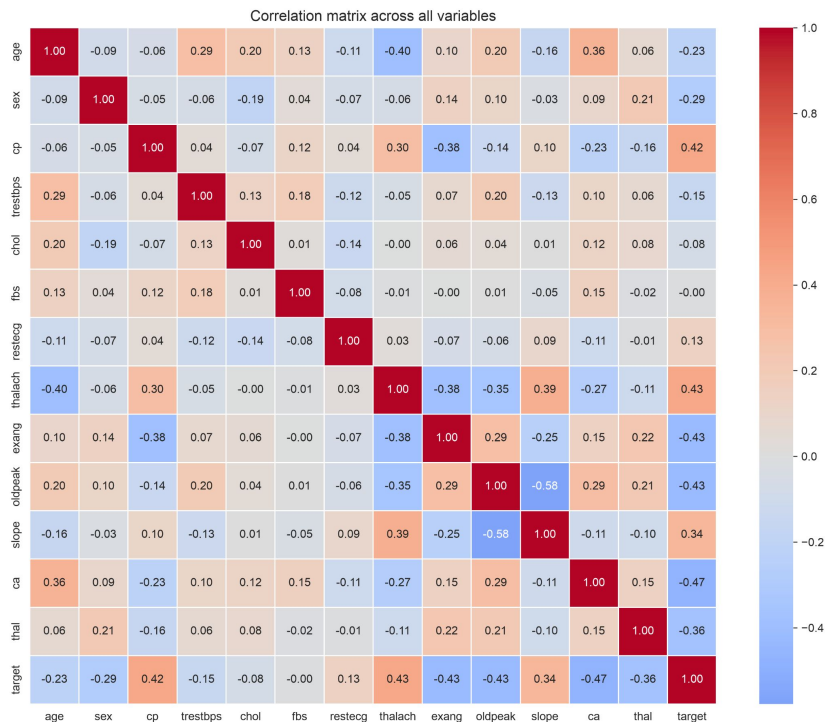
Target Distribution



- Nearly equal class split (54% diseased / 46% healthy)
- No class imbalance → accuracy is reliable
 - Exemple : 95/5 split where predicting "healthy" for everyone would give 95% accuracy without detecting a single sick patient

Exploratory Data Analysis

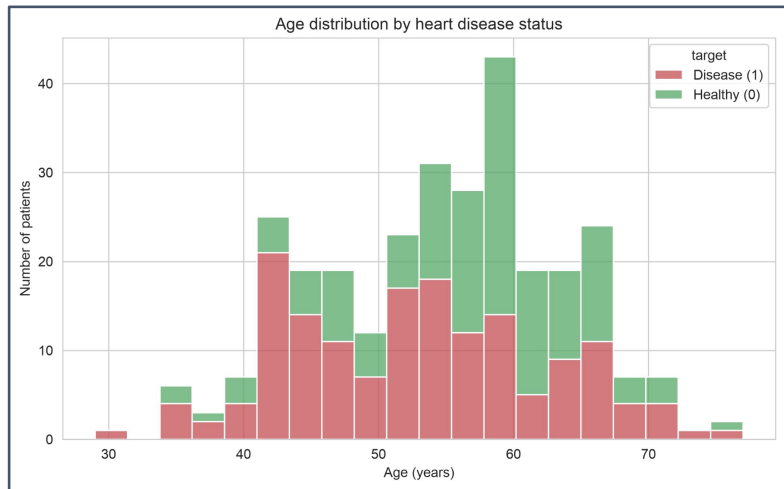
Key Correlations with Target



Strongest predictors identified via Pearson correlation:

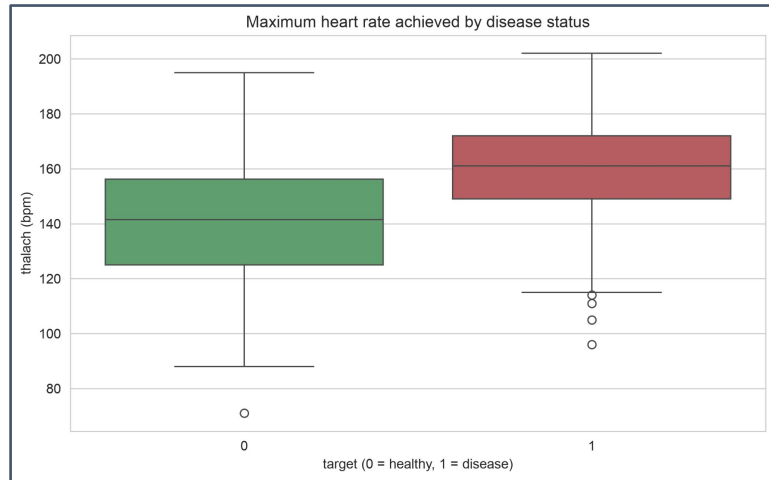
- Chest pain type (*cp*)
- Maximum heart rate (*thalach*)
- Exercise-induced angina (*exang*)
- ST depression (*oldpeak*)

Visual Insights



Age Distribution

Most patients are 45–65, peaking around 55 — consistent with cardiovascular epidemiology.



Max Heart Rate by Class

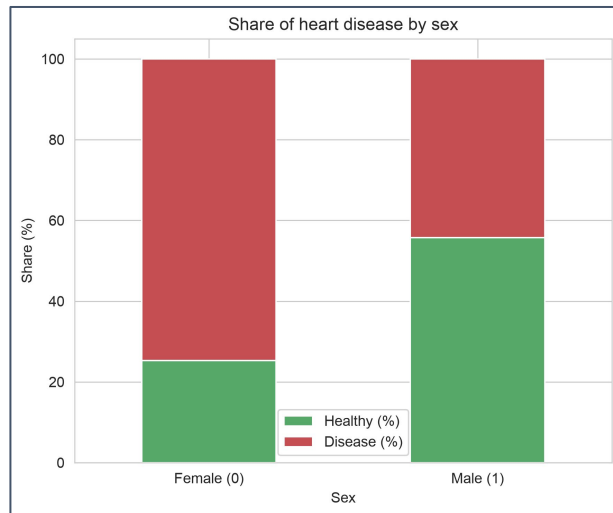
Diseased patients achieve higher max heart rates — a clear separation signal between classes.

Visual Insights



Chest Pain Type

Some chest pain categories are far more associated with disease — strong predictive power expected.



Heart disease by sex

Disease share is higher among women than men — pattern should be interpreted carefully given the small sample size.

Transforming Data and Features

One-Hot Encoding

	cp (values:[0,1,2,3])
patient 1	0
patient 2	2



transform into
binary columns

	cp_1	cp_2	cp_3
patient 1	0	0	0
patient 2	0	1	0

1

avoid unnecessary classification that could harm the precision of models using **distance** and **order** in their calculations

2

Applied to the fields : **cp**, **restecg**, **slope**, **thal**

3

Dataset expanded from 14 to 19 columns.

Transforming Data and Features

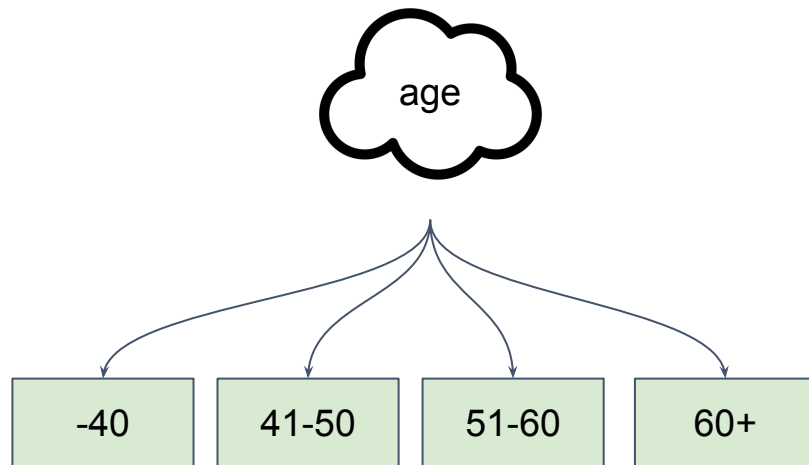
Standardisation

	initial range
age	0 - 99
oldpeak	0 - 6
chol	126 - 560
thalach	88 - 220

➡ We need to put the different columns on the same scale !

➡ Scaler used only on the train data set

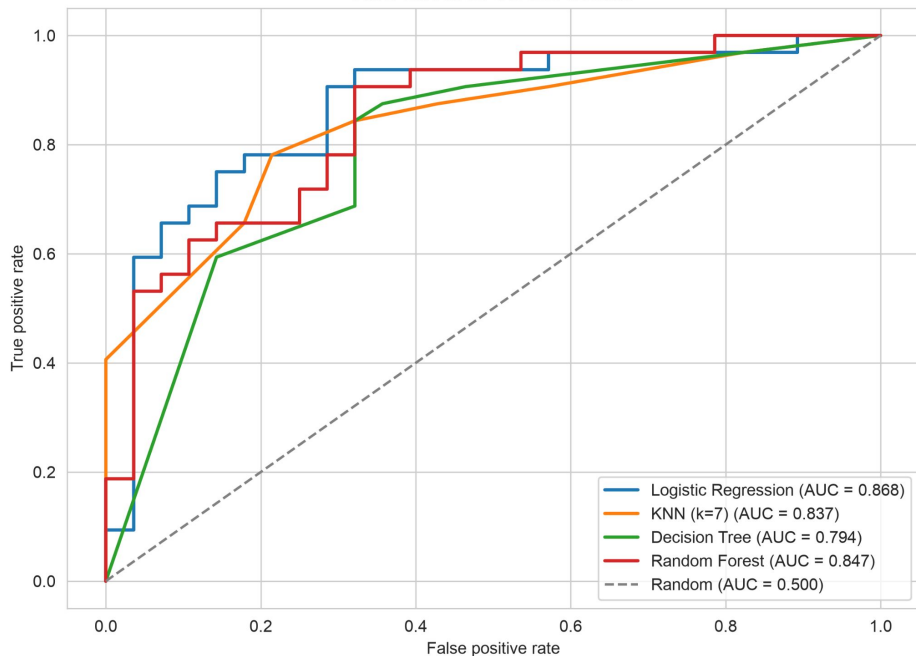
Data grouping



➡ For observation and classification purposes

Four Classification Models Compared

ROC curves for the four models



Model Selection & Training

- 80/20 split (236 train / 60 test)
- We gave standardized data to scale-sensitive models, and raw features to tree-based models

- **Logistic Regression** - interpretable linear baseline
- **KNN (k=7)** - instance-based classification
- **Decision Tree (max_depth=5)** - rule-based, interpretable
- **Random Forest (200 trees)** - a robust ensemble



Logistic Regression won ! highest F1-score (0.824) and recall (0.875). Simplicity outperformed complexity on this small dataset.

Evaluation

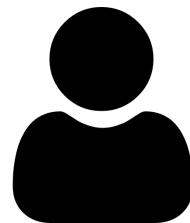
Test Set Performance (60 patients)



All models substantially
outperform random chance
(AUC > 0.79)



None of the models are
perfect



Small test set => only a
few patients difference
between models

LogReg F1

0.824

LogReg Recall

0.875

LogReg AUC

0.868

Ethical Considerations



Data leakage avoided

Duplicates removed before any split => honest F1 $\approx$ 0.82



Data privacy

A wrong prediction can cause unnecessary anxiety => for learning purposes only, not a medical tool



Data privacy

Dataset from one clinic, more men than women => results may not apply to all populations



Data privacy

Real-world use would require patient consent

Key Takeaways & Future Directions

→ Main findings

- From 1,025 raw rows to 296 clean patients
- Logistic Regression performed best
- Strongest predictor : chest pain type, ST depression and maximum heart rate

→ What we learned

- Always check for duplicates before modeling ⇒ high accuracy = warning sign
- A simple model can outperform complex ones on a small dataset.

→ Limitations & future work

- Only 296 patients from one clinic
- Larger and more diverse data would be needed to build a model usable in practice

